

Great work!

K	10 / 10	T	13 / 14	C	10 / 10	A	7 / 8
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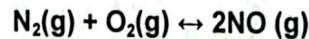
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Chapter 7 Quiz

Knowledge (10 marks) – Multiple Choice on D2L

Thinking (14 marks)

1. Consider the following reaction:



$$K = 0.088 \text{ at } 3727^\circ\text{C}$$

A reaction mixture at 3727°C initially contains 1.00 mol of nitrogen gas and 1.00 mol oxygen gas in a 2.00 L flask. Find the concentrations of all reactants and products at equilibrium at this temperature. (5 marks)

rule of 100

sig fig

4

$$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$$

I	0.5	0.5	0
C	-x	-x	+2x
E	0.5-x	0.5-x	2x

$C = \frac{n}{V}$
 $[\text{N}_2] = [\text{O}_2] = C = \frac{0.100 \text{ mol}}{1 \text{ L}} = 0.1 \text{ mol/L}$

$K = 0.088$

$K = \frac{[\text{products}]}{[\text{reactants}]}$

$K = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$

$0.088 = \frac{(2x)^2}{(0.5-x)(0.5-x)}$

rule of 100: $\frac{0.5}{0.088} < 100$
 doesn't apply

$\sqrt{0.088} = \frac{(2x)^2}{(0.5-x)^2}$

$0.2966(0.5-x) = 2x$

$0.1483 = 2x + 0.2966x$

$0.1483 = 2.2966x$

$x = 6.458 \times 10^{-2}$

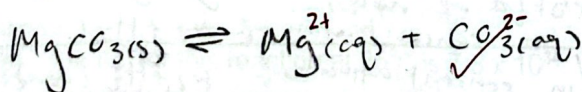
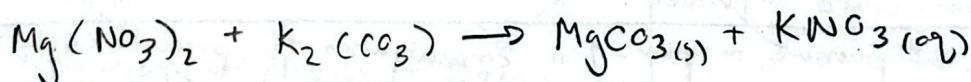
$x = 0.0646$

$[\text{N}_2]_{\text{eq}} = 0.435 \text{ mol/L}$

$[\text{O}_2]_{\text{eq}} = 0.435 \text{ mol/L}$

$[\text{NO}]_{\text{eq}} = 0.129 \text{ mol/L}$

2. Will a precipitate form if 40 mL of 0.0080 mol/L magnesium nitrate solution is added to 60 mL of a 0.010 mol/L solution of potassium carbonate? (5 marks)



$$[\text{Mg}] C_1 V_1 = C_2 V_2$$

$$C_2 = \frac{(0.04 \text{ L})(0.0080 \text{ mol/L})}{(0.04 + 0.06 \text{ L})}$$

$$= 3.2 \times 10^{-3} \text{ mol/L}$$

$$[\text{CO}_3] C_1 V_1 = C_2 V_2$$

$$C_2 = \frac{(0.06 \text{ L})(0.010 \text{ mol/L})}{(0.04 + 0.06 \text{ L})}$$

$$= 6 \times 10^{-3} \text{ mol/L}$$

$$Q = [\text{Mg}][\text{CO}_3]$$

$$Q = (3.2 \times 10^{-3})(6 \times 10^{-3})$$

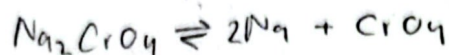
$$= 1.92 \times 10^{-5}$$

$$K_{\text{sp}} [\text{MgCO}_3] = 6.8 \times 10^{-6}$$

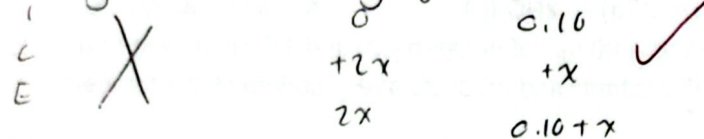
$Q > K_{\text{sp}}$ a precipitate will form

3. Calculate the molar solubility of solid silver chromate, $\text{Ag}_2\text{CrO}_4(\text{s})$, in 0.10 mol/L sodium chromate, $\text{Na}_2\text{CrO}_4(\text{aq})$. (4 marks)

common ion: CrO_4



1:1 mol ratio with CrO_4 : 0.10 mol/L



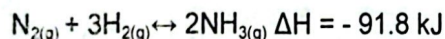
$$K_{sp} \text{ Ag}_2\text{CrO}_4: 1.1 \times 10^{-12}$$

100 rule: 0.10

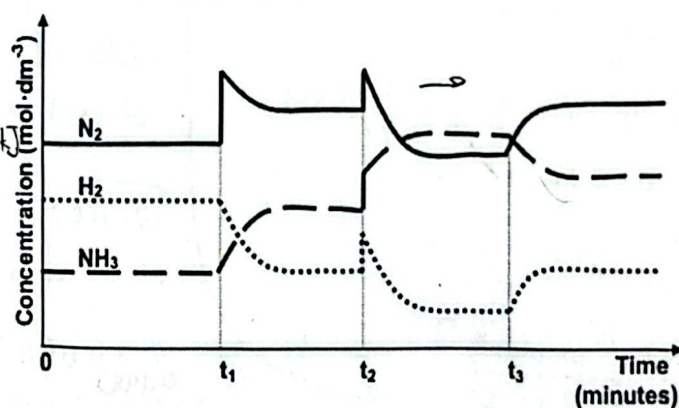
$\therefore$ applies $1.1 \times 10^{-12} > 100$

Communication (10 marks)

1. Identify the nature of the change imposed on the equilibrium system, shown in the graph below, at each of the times indicated t_1 , t_2 , t_3 .

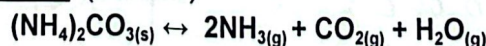


Also include the direction the equilibrium would shift to account for this change and state how the concentration of reactants and products change. Use point form to answer the question. (6 marks)



t_1	- increase in concentration of N_2 (reactant) - equilibrium shifts $\rightarrow$ products	- $\downarrow$ in concentration of $\text{N}_2 + \text{H}_2$ (reactants) - $\uparrow$ in concentration of NH_3 (product)
t_2	- increase of pressure in system - equil. shifts $\rightarrow$ products (least # of mols)	- $\uparrow$ in concentration of product (NH_3) - $\downarrow$ in concentration of $\text{N}_2 + \text{H}_2$ (reactant)
t_3	- increase in temperature - equil. shifts $\rightarrow$ towards reactants	- $\uparrow$ in concentration of $\text{N}_2 + \text{H}_2$ (reactant) - $\downarrow$ in concentration of NH_3 (product)

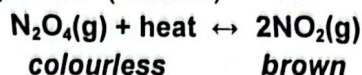
2. Suggest 2 different factors that could increase the production yield of ammonia gas in the reaction below. This reaction is endothermic. (2 marks)



- increase the concentration of $(\text{NH}_4)_2\text{CO}_3(\text{s})$ so equil. will be shifted towards products (ammonia) to consume additional reactant
- increase the temperature of the system, as it is endothermic energy is like a reactant, and equil. will shift $\rightarrow$ product to counteract the heat

SCH 4U1a

3. Explain why a syringe containing NO_2 gas will first get darker and then lighter in colour when compressed. Use the equilibrium equation: (2 marks)



shows how product concentration / reactant concentration $K_{eq} = \frac{[\text{NO}_2(\text{g})]^2}{[\text{N}_2\text{O}_4(\text{g})]}$

1. when compressing, the pressure increases, so there is a shift towards the side with the least mols (reactant). With the additional reactants, more is consumed to produce a higher concentration of brown gas. Also logically, more pressure = more intense color. As pressure increases, heat increases, resulting in greater conc. of NO_2 (brown). As gas escapes, pressure decreases, the color becomes lighter as volume increases and more $\text{N}_2\text{O}_4(\text{g})$ is produced.

Application (8 marks)

1. At a certain temperature, $K_{eq} = 0.18$ for the following equilibrium: $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$

Suppose a reaction vessel at this temperature contained these three gases at the following concentrations: $[\text{PCl}_3] = 0.0420 \text{ mol/L}$, $[\text{Cl}_2] = 0.0240 \text{ mol/L}$, $[\text{PCl}_5] = 0.00500 \text{ mol/L}$.

- a) Determine Q_{eq} for the system. (2 marks)
b) Is the system at equilibrium? If it is not, indicate the direction toward which the system will move to reach equilibrium. (1 mark)

a) $Q_{eq} = \frac{[\text{PCl}_5(\text{g})]}{[\text{PCl}_3(\text{g})][\text{Cl}_2(\text{g})]}$
 $= \frac{(0.00500)}{(0.0420)(0.0240)}$
 $= 4.96$ ✓

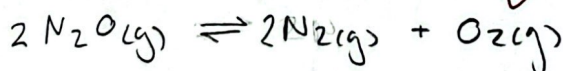
b) $K_{eq} = 0.18$

$Q_{eq} = 4.96$

$Q_{eq} > K_{eq}$ so the system is not @ equil. it must shift ← towards reactants to reach equil.

2. Dinitrogen monoxide (also known as laughing gas) is used as an anesthetic in dentistry. It can be made in many ways, but once made, it has a tendency to decompose into nitrogen and oxygen gas.

- a) Write the balanced equilibrium chemical equation for the above-mentioned reaction with states. (1 mark)



- b) The K_{eq} for this reaction at 25°C is 7.3×10^{34} . What can be said about the tendency of dinitrogen monoxide to decompose into nitrogen gas and oxygen gas? (1 mark)

because $K_{eq} = \frac{(\text{product concentration})}{(\text{reactant concentration})}$, the very high K_{eq} value indicates the concentration of products is much higher than reactants, showing how the system favours the forward reaction, so has a high tendency to decompose into $\text{N}_2 + \text{O}_2$.

3. For each example, complete the chart and predict the direction the equilibrium is shifted by the change imposed (L = left or R = right). If there is no shift, write NO SHIFT: (3 marks)

EQUATION	CHANGE IMPOSED	DIRECTION OF SHIFT
a) $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \leftrightarrow 2\text{CO}_2(\text{g})$	Volume is increased	← ✓
b) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \leftrightarrow 2\text{NH}_3(\text{g}) \quad \Delta H = -92 \text{ kJ}$	Temperature is decreased	→ ✓
c) $\text{PO}_4^{3-}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \leftrightarrow \text{HPO}_4^{2-}(\text{aq}) + \text{OH}^-(\text{aq})$	$\text{HCl}(\text{aq})$ is added	← NO shift.

if HCl acts as catalyst, no shift

OR if H^+ reacts with OH^- and H_2O so shift right